VIJAY KUMAR

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EDUCATION

B.TECH (Software Engineering)	2023-2027	Delhi Technological University, New Delhi	8.89
AISSCE/CBSE (Class XII)	2023	Jai Hind Public School	90.2 %
AISCE/CBSE (Class X)	2021	Kendriya Vidyalaya Pitampura	95.6 %

INTERNSHIPS

Google Summer of Code 2025 | OpenVINO Intel | Link — Remote | May 2025 - Aug 2025

- Reverse-engineered **MediaPipe's hand gesture recognition pipeline** in Python, implementing **custom SSD anchor decoding** and **geometric transformations** for OpenVINO optimization.
- Built **real-time gesture control system** with smart palm detection state machine, achieving **<16ms frame processing** and supporting multi-modal interactions.
- Engineered a **PyQt6-based dashboard** for gesture control with responsive layouts, dynamic modes (Browser, Media Player, PPT), and a custom mode builder **supporting 8+ configurable gesture mappings.**

ParkItUp | Intern | Link — Delhi, India | Feb 2025 – April 2025

- Improved a **parking web application** built with **Next.js** and **TypeScript**, reducing **load time by ~20–25%** through optimized routing and component-level rendering.
- Enhanced the app with role-based **dashboards for 4 user types**, adding access controls and streamlined UI flows that noticeably reduced task completion time.
- Optimized **Google Maps API** usage by implementing **caching**, **reducing latency** and **cutting redundant API calls**.

USIP Intern | HRDC, Delhi Technological University — Delhi, India | March 2025 – May 2025

- Improved HRDC DTU website with **React** and **lazy loading**, boosting load speed by 30%.
- Managed 20+ new updates, enhancing access for **500+ faculty and staff.**

ACADEMIC PROJECTS

PalmPilot – Adaptable Gesture Control Engine | Link | ***Python, OpenVINO, PyQt6, MediaPipe***

- **Engineered a high-performance gesture engine** by reverse-engineering MediaPipe's hand recognition pipeline and implementing custom **SSD anchor decoding** for OpenVINO optimization.
- Developed a **highly adaptable mapping system** that translates complex hand gestures into configurable OS-level events (Cursor, Keypress, Shortcuts) using **PyAutoGUI**.
- Designed a **PyQt6-based GUI** featuring a **Custom Mode Builder**, allowing users to create and save unique gesture profiles for software like PPT, Media Players, and Browsers.

InstaPrint– Digital Printing Management System | Link | ***NextJS, C#, Material UI, XAML, WPF***

- Built a **full-stack digital campus printing platform** that lets students submit print jobs remotely, select options, and track status in real time for 10+ users.
- Developed a **C# WPF/XAML desktop ops app** that auto-ingests new orders, applies print presets, and dispatches jobs to printers, cutting average processing time to **under 2 minutes**.
- **Synced order metadata and files** through a shared database/API layer, enabling operators to manage queues without back-and-forth, **reducing coordination overhead by ~60%.**

TECHNICAL SKILLS

Python, C++, JavaScript, TypeScript, C#, HTML, CSS, React.js, Next.js, Node.js, Tailwind CSS	OpenVINO, TensorFlow Lite, OpenCV, MediaPipe, Firebase, PyAutoGUI, WPF, XAML	Git, GitHub, Figma, MongoDB, MySQL

OPEN SOURCE

Contributor, OpenVINO Toolkit — Python, OpenVINO, PyTorch

- Submitted 3 pull requests (1 merged) to the PyTorch frontend, enhancing features and documentation through collaboration with maintainers. [Link]